q1_analysis

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Contents

Import and clean dataset

```
pancreatic_df <- read_csv("data/ebl.csv") |>
  janitor::clean_names() |>
  mutate(sex = case_when(sex == 0 ~ "Male",
                         sex == 1 ~ "Female"),
         sex = fct_relevel(sex, "Male"),
         race = case_when(race == 0 ~ "White",
                          race == 1 ~ "Black",
                          race == 2 ~ "Asian",
                          race == 3 ~ "Hispanic"),
         race = fct_relevel(race, "White"),
         op = as.factor(case_when(op == 0 ~ "Body or tail",
                                  op == 1 ~ "Whipple operation")),
         ven_res = as.factor(case_when(ven_res == 0 ~ "No",
                                       ven_res == 1 ~ "Yes")),
         neo_chemo = as.factor(case_when(neo_chemo == 0 ~ "No",
                                         neo_chemo == 1 ~ "Yes")),
         neo_xrt = as.factor(case_when(neo_xrt == 0 ~ "No",
                                       neo_xrt == 1 ~ "Yes")))
```

```
## Rows: 356 Columns: 13
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (13): patid, age, sex, race, liver_test, op, ebl, ven_res, neo_chemo, ne...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#head(pancreatic_df)
```

table 1: demographic characteristics + others a) Grouped by Sex

```
table1  <- pancreatic_df %>%
  dplyr::select(age:or_time) %>%
  tbl_summary(
```

```
    by = sex,
    statistic = list(
      all_continuous()  ~ "{mean} ({sd})",    # drop min/max to save space
      all_categorical() ~ "{n} ({p}%)"),
    label = list(age ~ "Age (years)",
      race ~ "Race",
      liver_test ~ "Bilirubin Level (in mg/dL)",
      op ~ "Type of Operation",
      ebl ~ "Estimated Intraoperative Blood Loss (in mL)",
      ven_res ~ "Resection + Reconstruction of Major Vasculature",
      neo_chemo ~ "Chemotherapy Prior to Surgery",
      neo_xrt ~ "Radiation Therapy Prior to Surgery",
      les_size ~ "Tumor Size (in cm)",
      or_time ~ "Operation Time (in mins)")) %>%
  add_p() %>%
  add_overall() %>%
  bold_labels() %>%
  modify_caption("**Baseline Characteristics by Sex**")
table1v2 <- table1 %>%
  as_gt() %>%
  gt::tab_options(
    data_row.padding = px(2),
    table.width      = pct(100),
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",    # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  cols_width(
    label      ~ px(150),
    everything() ~ px(80)
  ) |>
  gt::tab_style(
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything()))  |>
  gt::gtsave(filename = "figures_tables/table1.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmp6kWG5S/file11f92a81f07c.html screensho
```

```
system("pdfcrop figures_tables/table1.pdf figures_tables/table1.pdf")

table1 |>
  as_kable()
```

Table 1: **Baseline Characteristics by Sex**

Characteristic	**Overall** N = 356	**Male** N = 157	**Female** N = 199	**p-value**
Age (years)	63 (15)	66 (14)	61 (16)	0.011
Race				<0.001
White	217 (73%)	120 (90%)	97 (59%)	
Asian	9 (3.0%)	5 (3.8%)	4 (2.4%)	
Black	22 (7.4%)	3 (2.3%)	19 (12%)	
Hispanic	49 (16%)	5 (3.8%)	44 (27%)	
Unknown	59	24	35	
Bilirubin Level (in mg/dL)	5.3 (3.6)	5.8 (3.9)	4.8 (3.3)	0.020
Type of Operation				0.035
Body or tail	172 (48%)	66 (42%)	106 (53%)	
Whipple operation	184 (52%)	91 (58%)	93 (47%)	
Estimated Intraoperative Blood Loss (in mL)	658 (92)	675 (89)	644 (93)	0.002
Resection + Reconstruction of Major Vasculature	79 (22%)	47 (30%)	32 (16%)	0.002
Chemotherapy Prior to Surgery	67 (19%)	32 (20%)	35 (18%)	0.5
Radiation Therapy Prior to Surgery	39 (11%)	20 (13%)	19 (9.5%)	0.3
Tumor Size (in cm)	3.28 (2.35)	2.86 (1.71)	3.60 (2.71)	0.043
Operation Time (in mins)	374 (134)	404 (140)	351 (125)	<0.001

b) Grouped by Race

```
table1byrace  <- pancreatic_df %>%
  dplyr::select(age:or_time) %>%
  tbl_summary(
    by = race,
    statistic = list(all_continuous()  ~ "{mean} ({sd})",    # drop min/max to save space
      all_categorical() ~ "{n} ({p}%)"),
    label = list(age ~ "Age (years)",
      sex ~ "Sex",
      liver_test ~ "Bilirubin Level (in mg/dL)",
      op ~ "Type of Operation",
      ebl ~ "Estimated Intraoperative Blood Loss (in mL)",
      ven_res ~ "Resection + Reconstruction of Major Vasculature",
      neo_chemo ~ "Chemotherapy Prior to Surgery",
      neo_xrt ~ "Radiation Therapy Prior to Surgery",
      les_size ~ "Tumor Size (in cm)",
      or_time ~ "Operation Time (in mins)")) %>%
  add_p() %>%
  add_overall() %>%
  bold_labels() %>%
  modify_caption("**Baseline Characteristics by Race**")
```

```
## 59 missing rows in the "race" column have been removed.
```

```
table1bv2 <- table1byrace %>%
  as_gt() %>%
  gt::tab_options(
    data_row.padding = px(2),
    table.width      = pct(100),
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",     # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  cols_width(
    label       ~ px(150),
    everything() ~ px(80)
  ) |>
  gt::tab_style(
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything()))  |>
  gt::gtsave(filename = "figures_tables/table1b.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmp6kWG5S/file11f924c6f11e5.html screensh
```

```
system("pdfcrop figures_tables/table1b.pdf figures_tables/table1b.pdf")

table1byrace |>
  as_kable()
```

Table 2: **Baseline Characteristics by Race**

Characteristic	**Overall** N = 297	**White** N = 217	**Asian** N = 9	**Black** N = 22	**Hispanic** N = 49	**p-value**
Age (years)	63 (15)	64 (15)	63 (9)	61 (18)	63 (16)	0.9
Sex						<0.001
Male	133 (45%)	120 (55%)	5 (56%)	3 (14%)	5 (10%)	
Female	164 (55%)	97 (45%)	4 (44%)	19 (86%)	44 (90%)	
Bilirubin Level (in mg/dL)	5.4 (3.7)	5.8 (3.7)	1.9 (2.1)	5.6 (4.8)	4.0 (2.7)	<0.001
Type of Operation						0.4
Body or tail	144 (48%)	109 (50%)	2 (22%)	11 (50%)	22 (45%)	
Whipple operation	153 (52%)	108 (50%)	7 (78%)	11 (50%)	27 (55%)	
Estimated Intraoperative Blood Loss (in mL)	655 (94)	657 (100)	658 (106)	680 (92)	638 (51)	0.2
Resection + Reconstruction of Major Vasculature	61 (21%)	54 (25%)	2 (22%)	4 (18%)	1 (2.0%)	<0.001

Characteristic	**Overall** N = 297	**White** N = 217	**Asian** N = 9	**Black** N = 22	**Hispanic** N = 49	**p-value**
Chemotherapy Prior to Surgery	57 (19%)	52 (24%)	0 (0%)	4 (18%)	1 (2.0%)	<0.001
Radiation Therapy Prior to Surgery	30 (10%)	30 (14%)	0 (0%)	0 (0%)	0 (0%)	0.003
Tumor Size (in cm)	3.29 (2.40)	2.98 (1.75)	2.10 (0.74)	5.22 (3.81)	4.01 (3.49)	0.032
Operation Time (in mins)	373 (134)	378 (145)	386 (102)	418 (50)	328 (100)	0.006

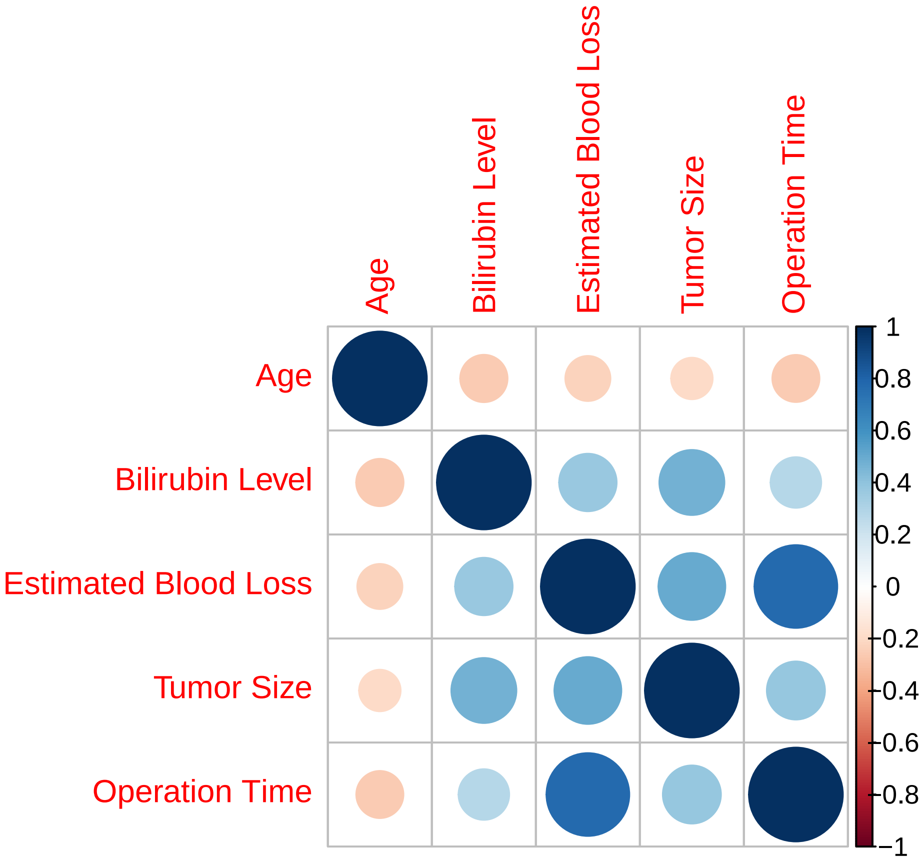
###correlation between predictors and within covariates

Male

```r
cor_dat <- pancreatic_df |>
  dplyr::filter(sex == "Male") |>
  dplyr::select(age, liver_test, ebl, les_size, or_time)
cor_matrix <- cor(cor_dat, method = "pearson")
rownames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Ti
colnames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Ti
pdf("figures_tables/corr_contvars_male.pdf", width = 6, height = 5)
corrplot(cor_matrix, method = "circle")
dev.off()
```

```
## pdf
##   2
```

```r
corrplot(cor_matrix, method = "circle")
```

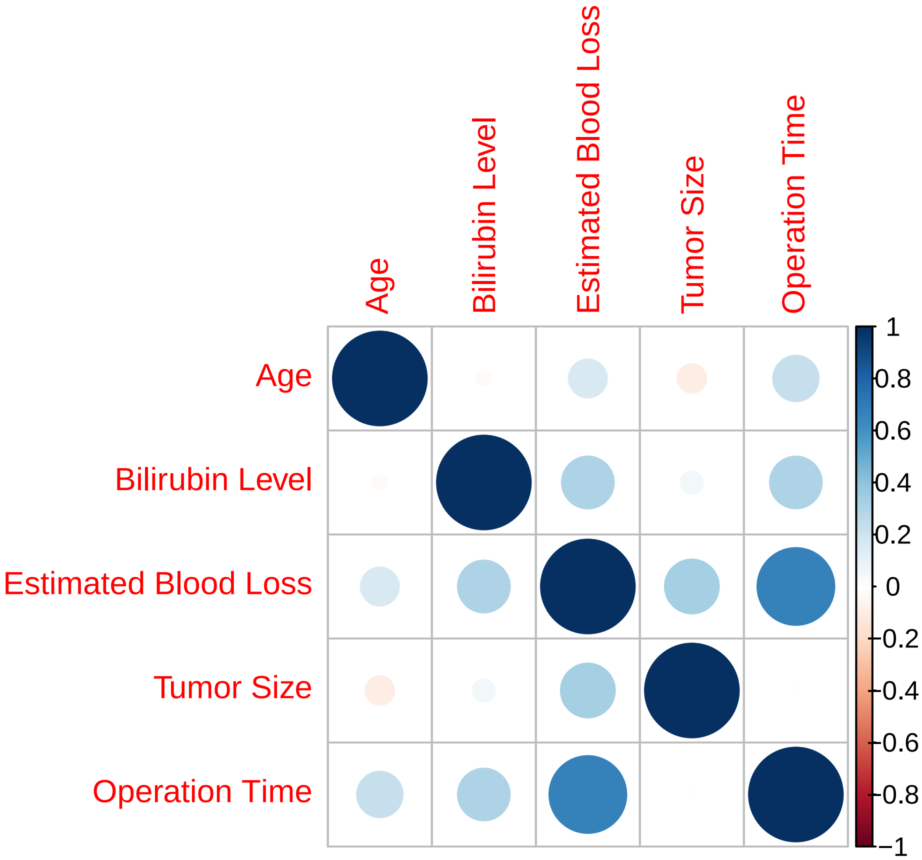


Female

```
cor_dat <- pancreatic_df |>
  dplyr::filter(sex == "Female") |>
  dplyr::select(age, liver_test, ebl, les_size, or_time)
cor_matrix <- cor(cor_dat, method = "pearson")
rownames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Ti
colnames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Ti
pdf("figures_tables/corr_contvars_female.pdf", width = 6, height = 5)
corrplot(cor_matrix, method = "circle")
dev.off()
```

```
## pdf
##   2
```

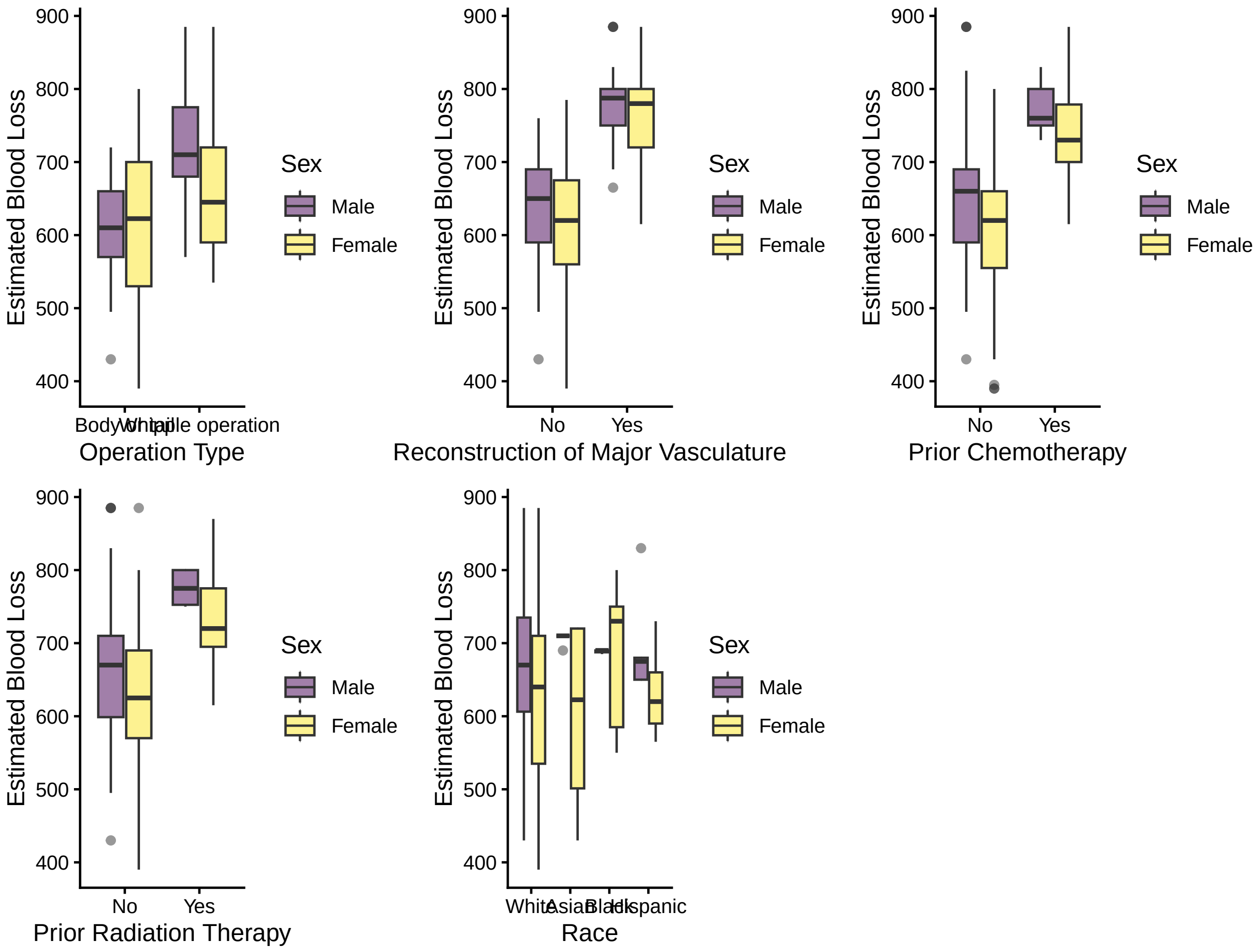
```
corrplot(cor_matrix, method = "circle")
```



operation time and age negatively correlated in men but no to low positive correlation in women

EBL Distribution in each categorical covariates

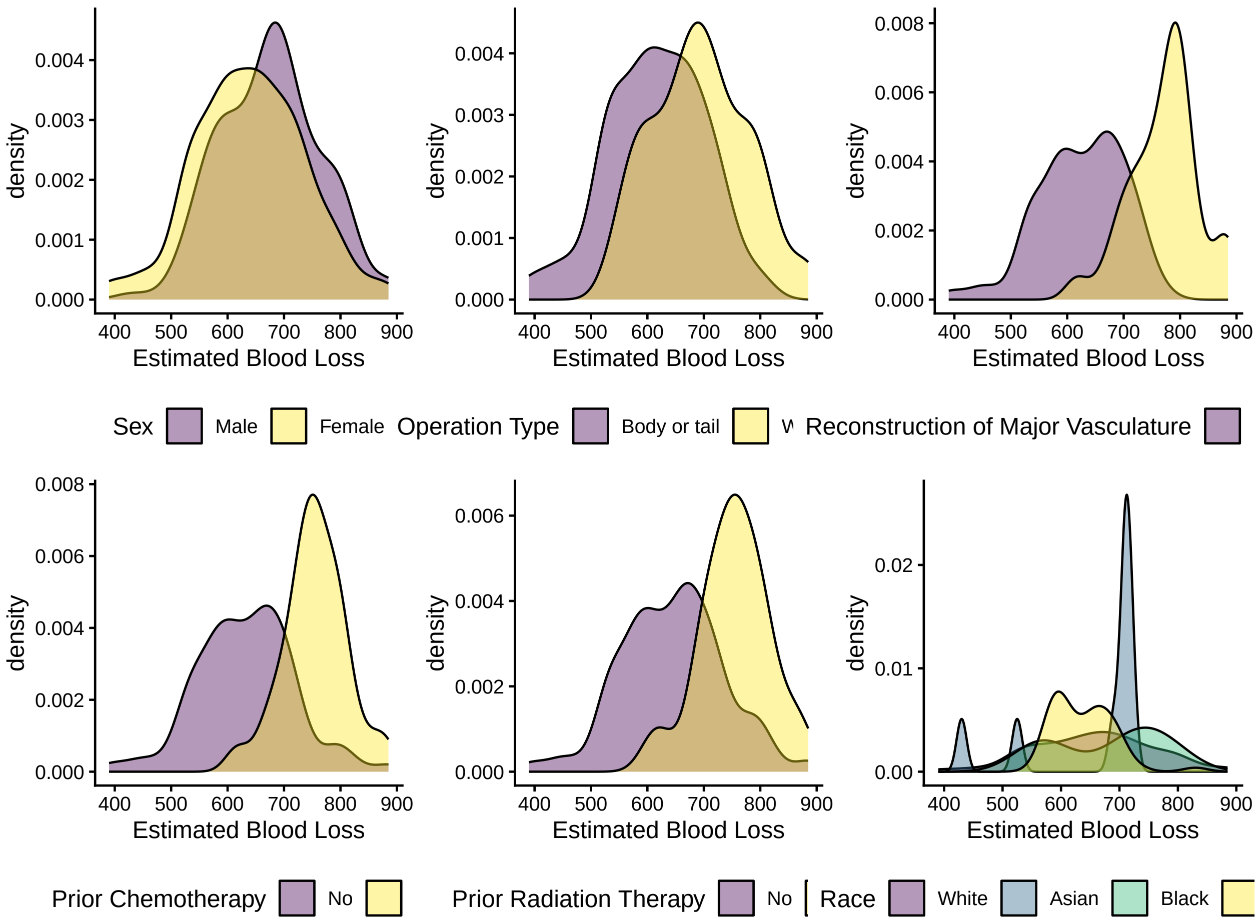
```r
colnames <- c("op","ven_res", "neo_chemo", "neo_xrt", "race")
names <- c("Operation Type", "Reconstruction of Major Vasculature", "Prior Chemotherapy", "Prior Radiat
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
                 aes(x = get(colnames[x]), y = ebl, fill = sex)) +
    geom_boxplot(alpha = 0.5) +
    labs (x = names[x], y = "Estimated Blood Loss",
          fill = "Sex") +
    theme_classic()
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
```




```
pdf("figures_tables/ebl_dist_across_cat_boxplot.pdf",
       width = 8, height = 7)
do.call(gridExtra::grid.arrange, c(plots, ncol = 2))
dev.off()
```

```
## pdf
##   2
```

```
colnames <- c("sex", "op","ven_res", "neo_chemo", "neo_xrt", "race")
names <- c("Sex", "Operation Type", "Reconstruction of Major Vasculature", "Prior Chemotherapy", "Prior
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
                aes(x = ebl, fill = get(colnames[x]))) +
    geom_density(alpha = 0.4) +
    labs (fill = names[x], x = "Estimated Blood Loss") +
    theme_classic() +
    theme(legend.position = "bottom")
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
```




```
pdf("figures_tables/ebl_dist_across_cat_densityplot.pdf",
      width = 8, height = 7)
do.call(gridExtra::grid.arrange, c(plots, ncol = 2))
dev.off()
```

```
## pdf
##   2
```

EBL lower in female regardless of prior chemotherapy, radiation therapy and major reconstruction status. However, EBL in female vs male are different depending on operation type (effect modification probably, can try interaction term). Low EBL in white and hispanic women. Very few asian and black men recruited in the study so comparison is difficult.

EBL's distribution is relatively normal across each group.

Recode race:

```
pancreatic_df <- pancreatic_df |>
  mutate(race = case_when(race == "White" ~ "White",
                          race %in% c("Asian", "Black", "Hispanic") ~ "Non-white"),
         race = fct_relevel(race, "White"))
```

Modeling:

```r
lm_v0 <- lm(ebl ~ race + sex + age +  sex + op + ven_res + neo_chemo +
              neo_xrt + liver_test + les_size + age*or_time,
            data = pancreatic_df)

summary(lm_v0)
```

```
##
## Call:
## lm(formula = ebl ~ race + sex + age + sex + op + ven_res + neo_chemo +
##     neo_xrt + liver_test + les_size + age * or_time, data = pancreatic_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -188.618  -24.731   -4.808   37.847  184.608
##
## Coefficients:
##                        Estimate Std. Error t value           Pr(>|t|)
## (Intercept)          368.432363  46.088491   7.994 0.0000000000000328 ***
## raceNon-white          9.082931   8.491519   1.070            0.28568
## sexFemale            -21.894965   7.301475  -2.999            0.00295 **
## age                    1.848933   0.726246   2.546            0.01143 *
## opWhipple operation    7.733185   8.968781   0.862            0.38928
## ven_resYes            26.734941  12.628408   2.117            0.03512 *
## neo_chemoYes          57.478235  12.641607   4.547 0.0000080633704538 ***
## neo_xrtYes           -41.818401  15.999351  -2.614            0.00943 **
## liver_test             1.738283   1.001478   1.736            0.08369 .
## les_size              10.771707   1.527664   7.051 0.0000000000134454 ***
## or_time                0.585633   0.120583   4.857 0.0000019714221246 ***
## age:or_time           -0.004202   0.001862  -2.257            0.02477 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 54.91 on 285 degrees of freedom
##   (59 observations deleted due to missingness)
## Multiple R-squared:  0.6697, Adjusted R-squared:  0.6569
## F-statistic: 52.53 on 11 and 285 DF,  p-value: < 0.00000000000000022
```

Type of operation not really significant

```r
lm_v1 <- lm(ebl ~ race + sex + age +  sex + ven_res + neo_chemo +
              neo_xrt + liver_test + les_size + age*or_time,
            data = pancreatic_df)

anova(lm_v1, lm_v0)
```

```
## Analysis of Variance Table
##
## Model 1: ebl ~ race + sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     liver_test + les_size + age * or_time
## Model 2: ebl ~ race + sex + age + sex + op + ven_res + neo_chemo + neo_xrt +
##     liver_test + les_size + age * or_time
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
```

```
## 1    286 861518
## 2    285 859277  1    2241.5 0.7434 0.3893
```

the model with operation time doesn't provide better fit. So removing that

```
lm_v3 <- lm(ebl ~ race+sex + age +  sex + ven_res + neo_chemo +
              neo_xrt + liver_test + les_size + age + or_time,
            data = pancreatic_df)
anova(lm_v3,lm_v1)
```

```
## Analysis of Variance Table
##
## Model 1: ebl ~ race + sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     liver_test + les_size + age + or_time
## Model 2: ebl ~ race + sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     liver_test + les_size + age * or_time
##   Res.Df    RSS Df Sum of Sq     F  Pr(>F)
## 1    287 877251
## 2    286 861518  1     15733 5.223 0.02302 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

One with interactions for raceXsex, liver_testXles_size, ageXor_time significantly better

```
#check_model(lm_v1)
#ggsave(filename = "figures_tables/lm_diagnostic_plot.pdf",
    #   width = 10, height = 15)
```

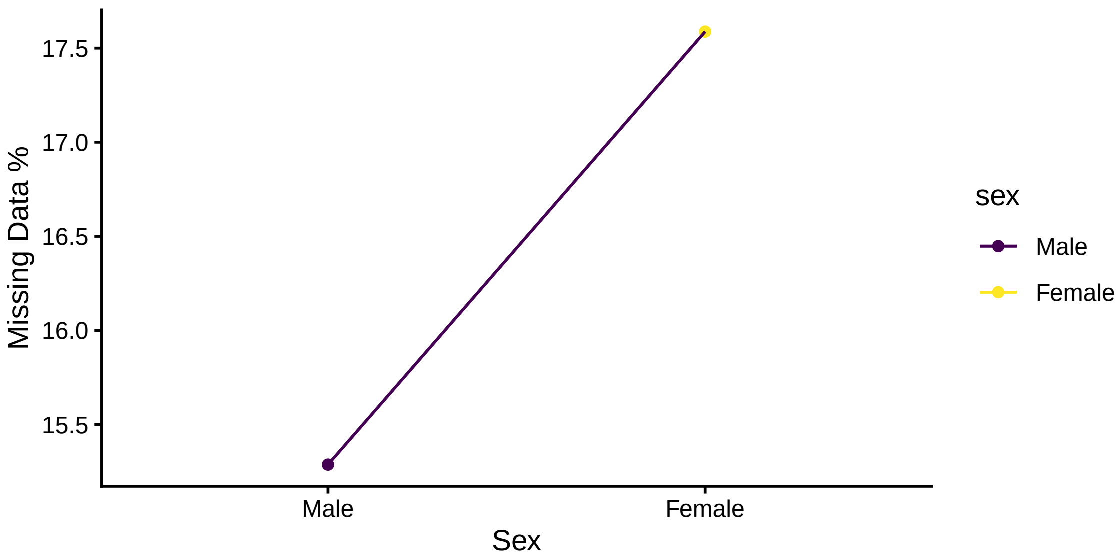
###Look at missingness

```
fig2b <- vis_miss(pancreatic_df |> dplyr::select(-prot_energy)) +
  theme(axis.text.x = element_text(vjust = 0))
ggsave("figures_tables/missingness_info.png", fig2b, height = 7, width =10)
```

Missingness by gender and age

```
race_gender_missing <- pancreatic_df |>
  dplyr::select(-prot_energy) |>
  group_by(sex) |>
  mutate(total = length(unique(patid))) |>
  ungroup() |>
  drop_na(race) |>
  group_by(sex) |>
  summarize (missing_pct = (total[1] - length(unique(patid)))*100/total[1]) |>
  mutate(dummy_group = 1)

missing_by_gender <- ggplot(aes(x = sex, y = missing_pct, group = dummy_group, color = sex),
       data = race_gender_missing) +
  geom_point() +
  geom_line() +
  geom_line() +
  labs(x = "Sex", y = "Missing Data %") +
```

```
  theme_classic()

ggsave("figures_tables/missingness_byracegender.pdf", missing_by_gender, height = 3, width =5)
system("pdfcrop figures_tables/missingness_byracegender.pdf figures_tables/missingness_byracegender.pdf
missing_by_gender
```



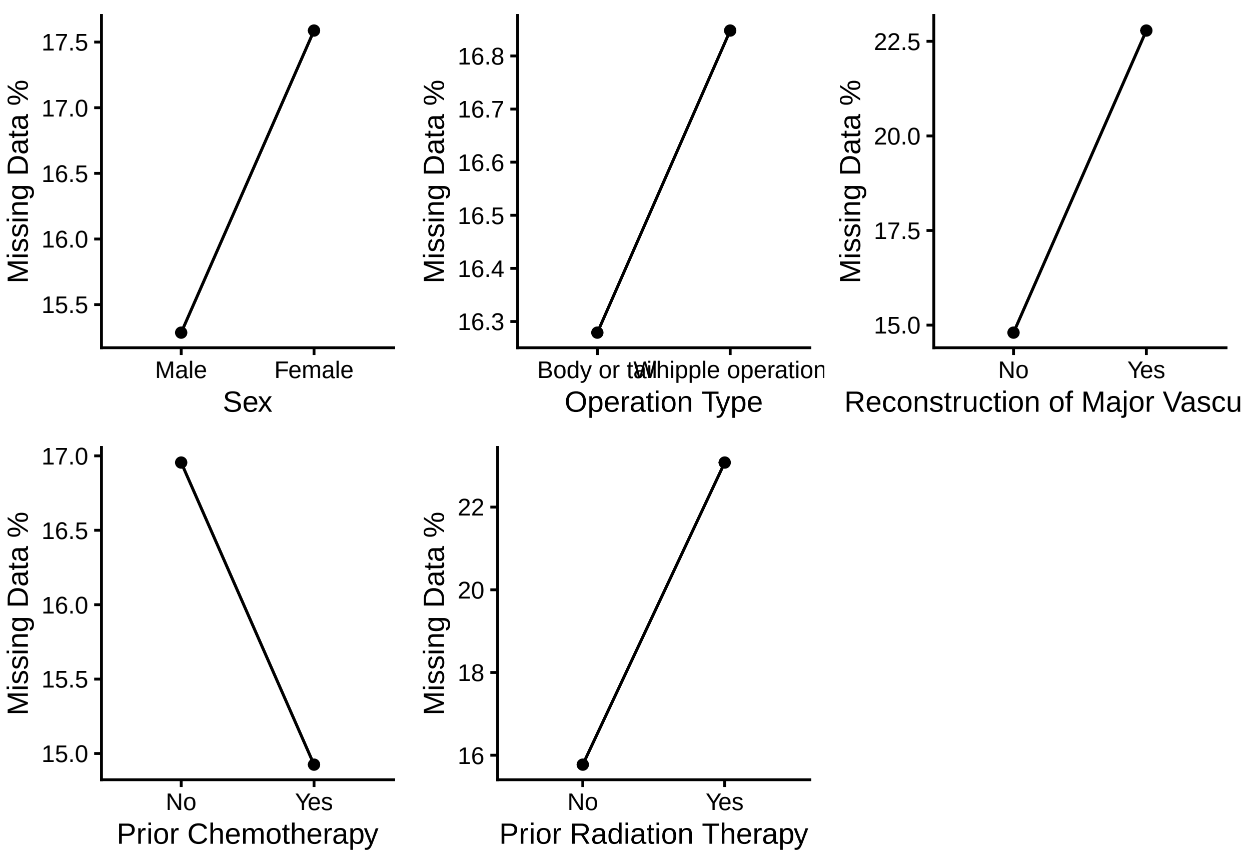
Look for other possible sources. - check trend of ebl across each continuous variable to see if its even linear or not, if not linear can try gams

Missingness in different quantile of age or other continuous variables

```
colnames <- c("sex", "op","ven_res", "neo_chemo", "neo_xrt")
names <- c("Sex", "Operation Type", "Reconstruction of Major Vasculature", "Prior Chemotherapy", "Prior
categorical_missing_pct <- lapply(1:length(colnames), function(x){
  race_gender_missing <- pancreatic_df |>
    dplyr::select(-prot_energy) |>
    group_by(get(colnames[x])) |>
    mutate(total = length(unique(patid))) |>
    ungroup() |>
    drop_na(race) |>
    group_by(get(colnames[x])) |>
    summarize (missing_pct = (total[1] - length(unique(patid)))*100/total[1]) |>
    mutate(dummy_group = 1)
  colnames(race_gender_missing) <- c(colnames[x], "missing_pct", "dummy_group")
  print(head(race_gender_missing))

  missing_by_cat <- ggplot(aes(x = get(colnames[x]), y = missing_pct, group = dummy_group),
         data = race_gender_missing) +
    geom_point() +
    geom_line() +
    geom_line() +
    labs(x = names[x], y = "Missing Data %") +
    theme_classic()

  return(missing_by_cat)
})
```

```
## # A tibble: 2 x 3
##   sex    missing_pct dummy_group
##   <fct>        <dbl>       <dbl>
## 1 Male          15.3           1
## 2 Female        17.6           1
## # A tibble: 2 x 3
##   op                 missing_pct dummy_group
##   <fct>                    <dbl>       <dbl>
## 1 Body or tail              16.3           1
## 2 Whipple operation         16.8           1
## # A tibble: 2 x 3
##   ven_res missing_pct dummy_group
##   <fct>         <dbl>       <dbl>
## 1 No             14.8           1
## 2 Yes            22.8           1
## # A tibble: 2 x 3
##   neo_chemo missing_pct dummy_group
##   <fct>           <dbl>       <dbl>
## 1 No               17.0           1
## 2 Yes              14.9           1
## # A tibble: 2 x 3
##   neo_xrt missing_pct dummy_group
##   <fct>         <dbl>       <dbl>
## 1 No             15.8           1
## 2 Yes            23.1           1
```

```r
do.call(gridExtra::grid.arrange, c(categorical_missing_pct, ncol = 3))
```



```
pdf("figures_tables/missingness_percategorical_group.pdf",
       width = 9, height = 8)
do.call(gridExtra::grid.arrange, c(categorical_missing_pct, ncol = 3))
dev.off()
```

```
## pdf
##   2
```

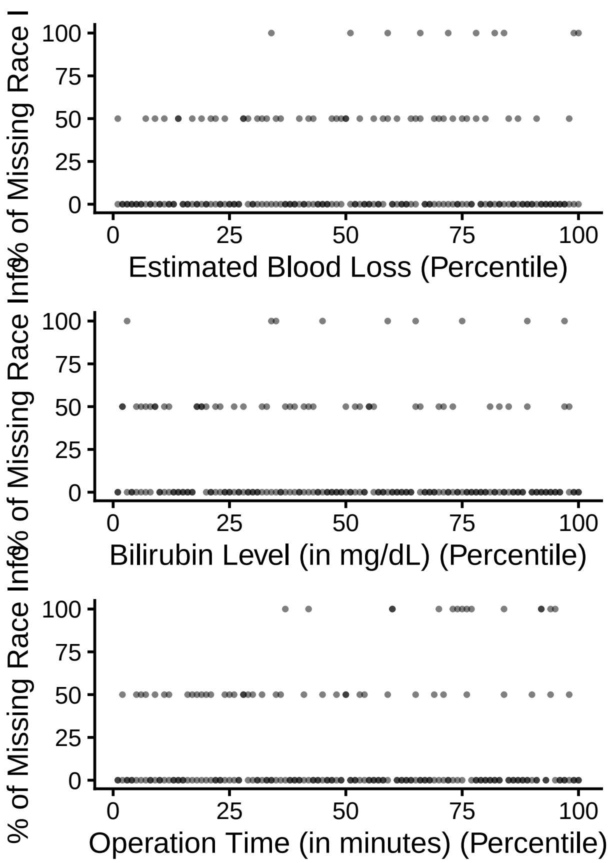
```
colnames <- c("ebl", "age","liver_test", "les_size", "or_time")
names <- c("Estimated Blood Loss", "Age", "Bilirubin Level (in mg/dL)", "Tumor Size (in cm)", "Operatio
get_quintile_missing_pct <- lapply(1:length(colnames), function(x){
  df_filter <- pancreatic_df %>%
    group_by(sex) |>
    mutate(xcol = get(colnames[x]),
           x_percentile = ntile(xcol, 100)) |>
    group_by(x_percentile, sex) |>
    summarise(n = n(),
              missing_y = sum(is.na(race)),
              group = paste(as.character(min(xcol)), as.character(max(xcol)), sep = "-"),
              covariate = colnames[x],
              pct_missing_y = mean(is.na(race)) * 100)

  plot <- ggplot(df_filter,
                 aes(x = x_percentile, y = pct_missing_y)) +
    geom_point(size = 0.5, alpha = 0.5) +
    labs (x = paste(names[x], "(Percentile)" ), y = "% of Missing Race Info",
          color = "Sex") +
  theme_classic() +
    theme(legend.position = "none")

  return(plot)
})
```

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by x_percentile and sex.
## i Output is grouped by x_percentile.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(x_percentile, sex))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
do.call(gridExtra::grid.arrange, c(get_quintile_missing_pct, ncol = 2))
```

```
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
```

```
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
```




```r
pdf("figures_tables/missingness_percontinuous_group.pdf",
       width = 10, height = 8)
do.call(gridExtra::grid.arrange, c(get_quintile_missing_pct, ncol = 2))
```

```
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
```

```r
dev.off()
```

```
## pdf
##   2
```

The distribution is missing percent against all the continous variables are randomly distributed.

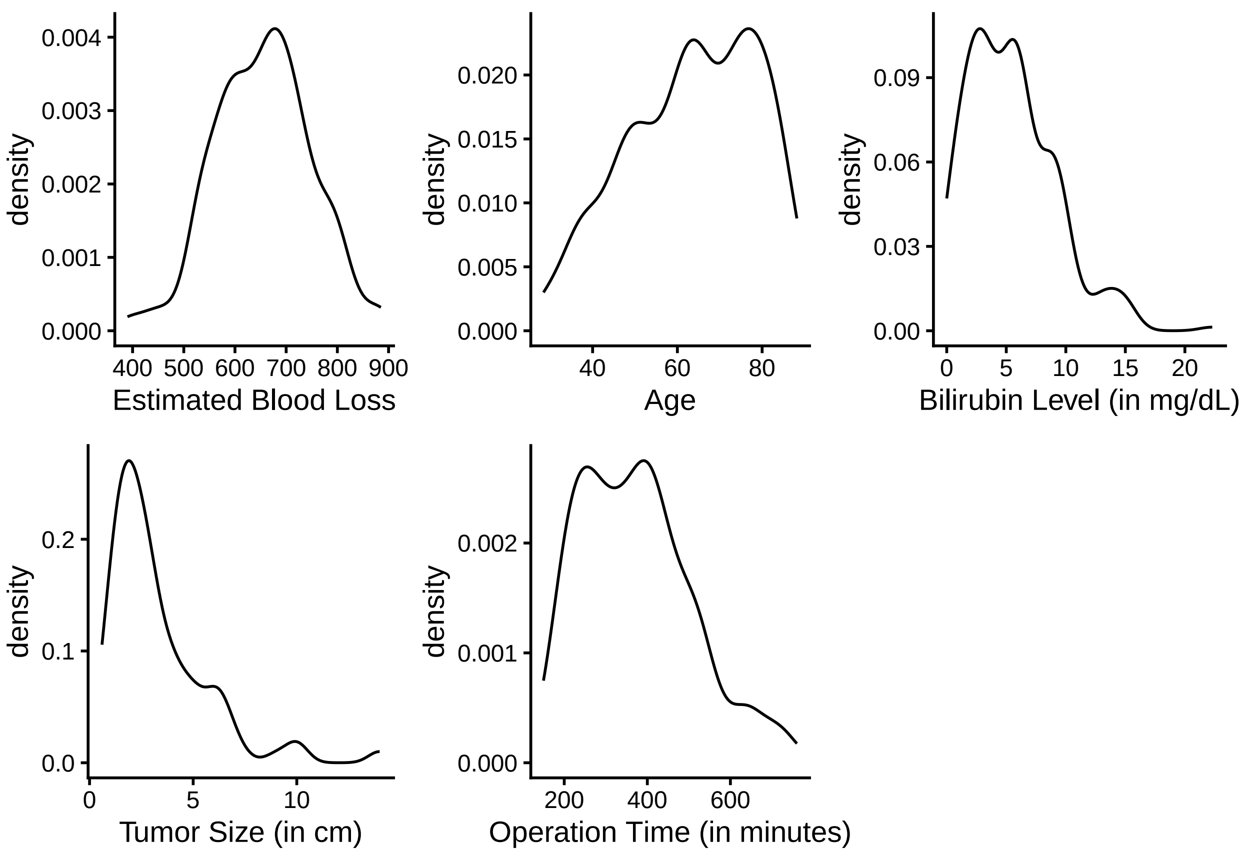
```
mcar_test(pancreatic_df |>  dplyr::select(-prot_energy))
```

```
## # A tibble: 1 x 4
##   statistic    df p.value missing.patterns
##       <dbl> <dbl>   <dbl>            <int>
## 1      18.9    11  0.0621                2
```

NS –> MCAR holds

Tumor size (heavily left-skewed)

```
colnames <- c("ebl","age","liver_test", "les_size", "or_time")
names <- c("Estimated Blood Loss", "Age", "Bilirubin Level (in mg/dL)", "Tumor Size (in cm)", "Operatio
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
                 aes(x = get(colnames[x]))) +
    geom_density(alpha = 0.5) +
    labs (x = names[x]) +
    theme_classic()
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
```

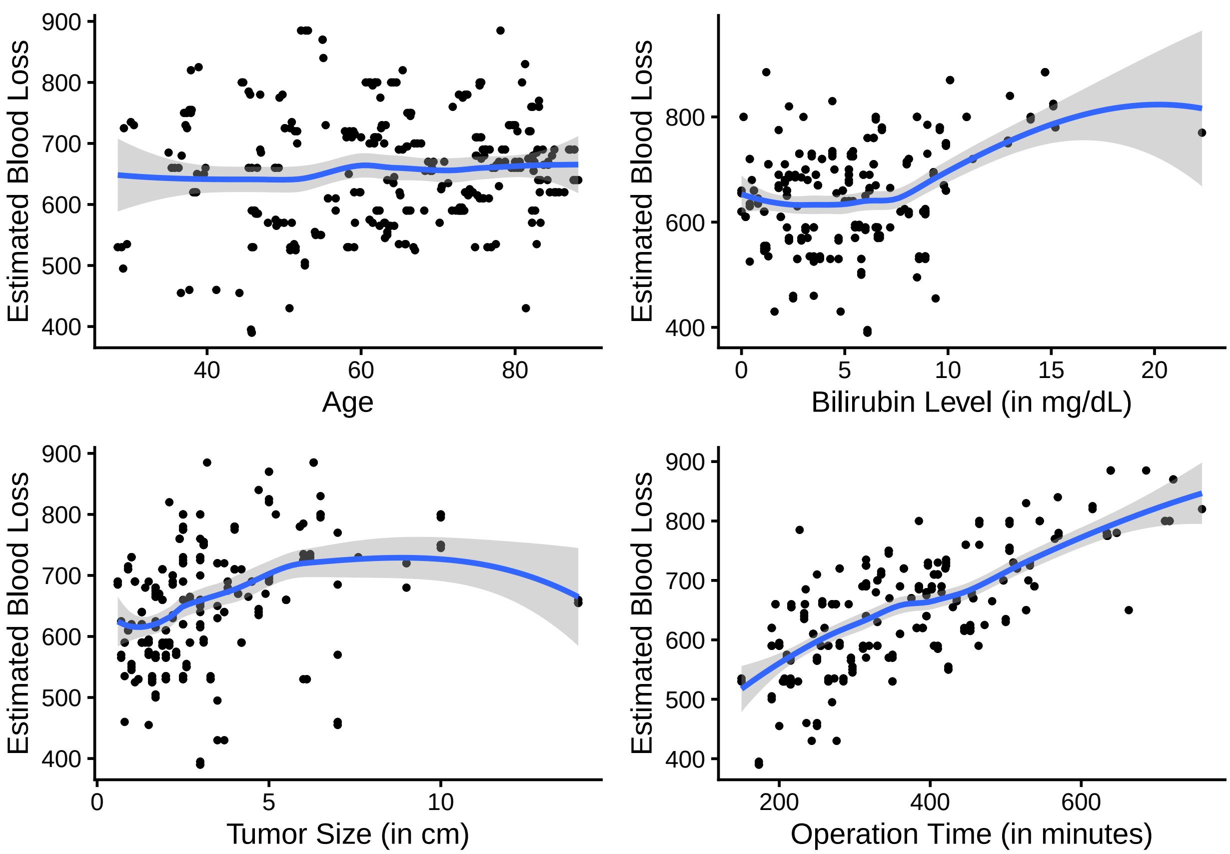


```
pdf("figures_tables/cont_variables_distribution.pdf",
       width = 8, height = 6)
do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
dev.off()
```

```
## pdf
##   2
```

```
colnames <- c("age","liver_test", "les_size", "or_time")
names <- c("Age", "Bilirubin Level (in mg/dL)", "Tumor Size (in cm)", "Operation Time (in minutes)")
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
                 aes(x = get(colnames[x]), y = ebl)) +
    geom_point(size = 0.8) +
    geom_smooth() +
    labs (x = names[x], y = "Estimated Blood Loss") +
    theme_classic()
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 2))
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



```r
pdf("figures_tables/ebl_versus_cont_variable.pdf",
       width = 8, height = 6)
do.call(gridExtra::grid.arrange, c(plots, ncol = 2))
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```r
dev.off()
```

```
## pdf
##   2
```

Bilirubin level - minimal association at smaller amount, increased association after 5

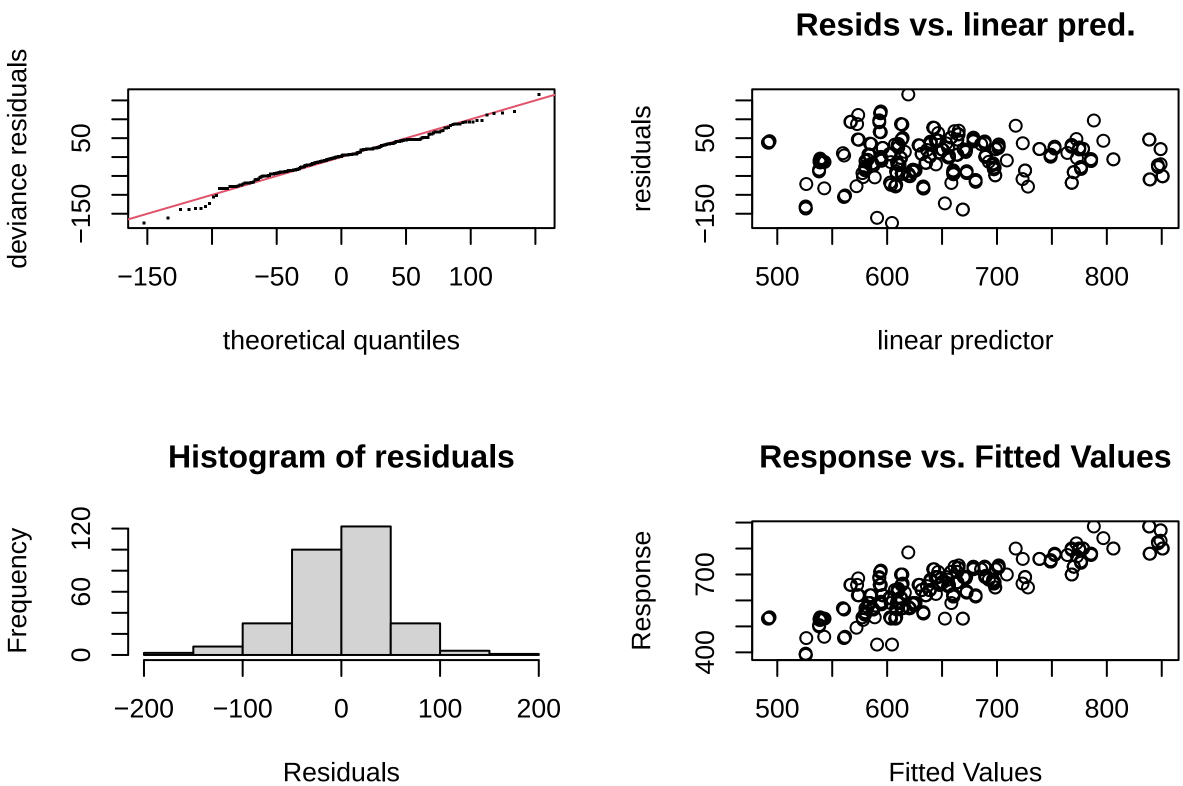
GAM

```r
gam_v1 <- gam(ebl ~ race + sex + age +  sex + ven_res + neo_chemo +
              neo_xrt + s(liver_test) + s(les_size) + age*or_time,
            data = pancreatic_df)

summary(gam_v1)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ebl ~ race + sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     s(liver_test) + s(les_size) + age * or_time
##
## Parametric coefficients:
##                   Estimate Std. Error t value             Pr(>|t|)
## (Intercept)     425.417254  46.337479   9.181 < 0.0000000000000002 ***
## raceNon-white    19.515210   8.581704   2.274               0.0237 *
## sexFemale       -29.275937   7.194435  -4.069         0.0000616832 ***
## age               1.668891   0.741682   2.250               0.0252 *
## ven_resYes       18.016577  12.732666   1.415               0.1582
## neo_chemoYes     72.200711  13.146637   5.492         0.0000000905 ***
## neo_xrtYes      -39.506290  15.934948  -2.479               0.0138 *
## or_time           0.522841   0.123706   4.226         0.0000323322 ***
## age:or_time      -0.003084   0.001946  -1.585               0.1142
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                   edf Ref.df     F             p-value
## s(liver_test) 4.758  5.762 0.912               0.421
## s(les_size)   8.279  8.832 9.856 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## R-sq.(adj) =  0.692   Deviance explained = 71.4%
## GCV = 2926.3  Scale est. = 2709.2    n = 297
```

```
gam.check(gam_v1)
```




```
##
## Method: GCV   Optimizer: magic
## Smoothing parameter selection converged after 11 iterations.
## The RMS GCV score gradient at convergence was 0.005103249 .
## The Hessian was positive definite.
## Model rank =  27 / 27
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                  k'  edf k-index              p-value
## s(liver_test) 9.00 4.76    0.69 <0.0000000000000002 ***
## s(les_size)   9.00 8.28    0.81 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
gam_v2 <- gam(ebl ~ race + sex + age +  sex + ven_res + neo_chemo +
              neo_xrt + s(liver_test) + s(les_size) + age+or_time,
            data = pancreatic_df)

summary(gam_v2)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ebl ~ race + sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     s(liver_test) + s(les_size) + age + or_time
##
## Parametric coefficients:
##                Estimate Std. Error t value             Pr(>|t|)
## (Intercept)   496.21672   18.53990  26.765 < 0.0000000000000002 ***
## raceNon-white  20.80744    8.67553   2.398               0.0171 *
## sexFemale     -32.15700    7.23439  -4.445         0.0000127884 ***
## age             0.51326    0.24812   2.069               0.0395 *
## ven_resYes     14.78113   12.88547   1.147               0.2523
## neo_chemoYes   72.75766   13.25265   5.490         0.0000000918 ***
## neo_xrtYes    -32.21246   16.34273  -1.971               0.0497 *
## or_time         0.33531    0.03731   8.986 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                 edf Ref.df     F             p-value
## s(liver_test) 7.628  8.504 1.709              0.0929 .
## s(les_size)   8.129  8.762 8.798 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.694   Deviance explained = 71.8%
## GCV = 2919.3  Scale est. = 2685.8    n = 297
```

```r
anova(gam_v1, gam_v2)
```

```
## Analysis of Deviance Table
##
## Model 1: ebl ~ race + sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     s(liver_test) + s(les_size) + age * or_time
## Model 2: ebl ~ race + sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     s(liver_test) + s(les_size) + age + or_time
##   Resid. Df Resid. Dev    Df Deviance      F  Pr(>F)
## 1    273.41     744936
## 2    271.73     733880 1.671    11056 2.4634 0.09675 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
pdf("figures_tables/gamv1_diagnostic_plot.pdf",
       width = 6, height = 5)
gam.check(gam_v2)
```

```
##
## Method: GCV   Optimizer: magic
## Smoothing parameter selection converged after 11 iterations.
## The RMS GCV score gradient at convergence was 0.03489012 .
```

```
## The Hessian was positive definite.
## Model rank =  26 / 26
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                 k'  edf k-index             p-value
## s(liver_test) 9.00 7.63    0.70 <0.0000000000000002 ***
## s(les_size)   9.00 8.13    0.81 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
dev.off()
```

```
## pdf
##   2
```

###Compare fit

```r
allmods_diag <- data.frame(
  Model = c("LM", "GAM_v1"),
  AIC   = c(AIC(lm_v1), AIC(gam_v2)),
  BIC   = c(BIC(lm_v1), BIC(gam_v2)),
  `log(Likelihood)` = c(as.numeric(logLik(lm_v1)), as.numeric(logLik(gam_v2))),
  DF     = c(attr(logLik(lm_v1), "df"),
             attr(logLik(gam_v2), "df"))
) |>
  kable(caption = "",
        booktabs = FALSE, longtable = FALSE, linesep = "", digits = 2, full_width = FALSE)  |>
  kable_styling(latex_options = "hold_position", font_size = 12)

#save_kable(allmods_diag, file = "figures_tables/model_comparison.pdf")
#system("pdfcrop figures_tables/model_comparison.pdf figures_tables/model_comparison.pdf")

allmods_diag
```

Table 3:

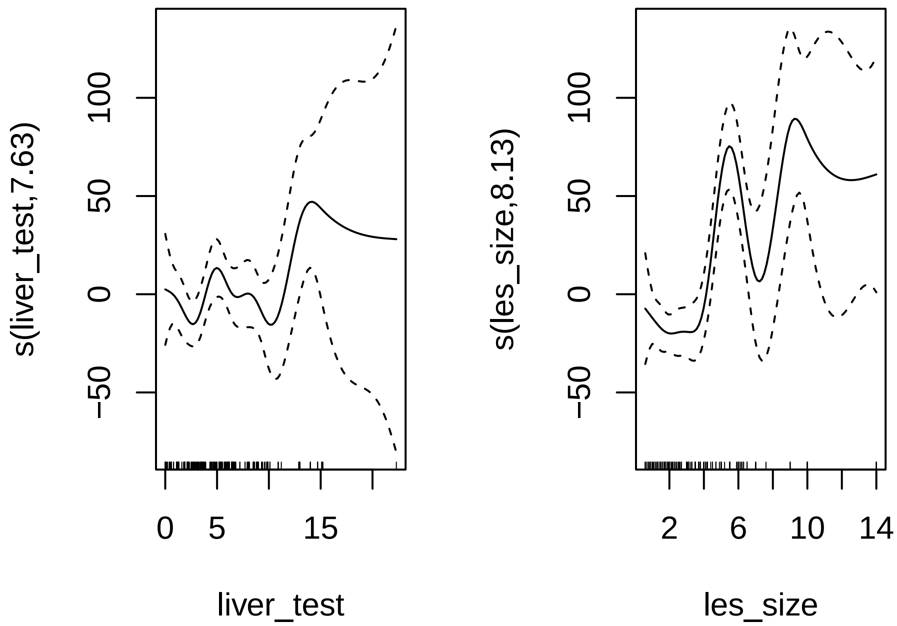
Model	AIC	BIC	log.Likelihood.	DF
LM	3234.75	3279.07	-1605.37	12.00
GAM_v1	3212.64	3304.08	-1581.56	24.76

Based on this, GAM_v1 is chosen (but BIC is higher, so need to look into this)

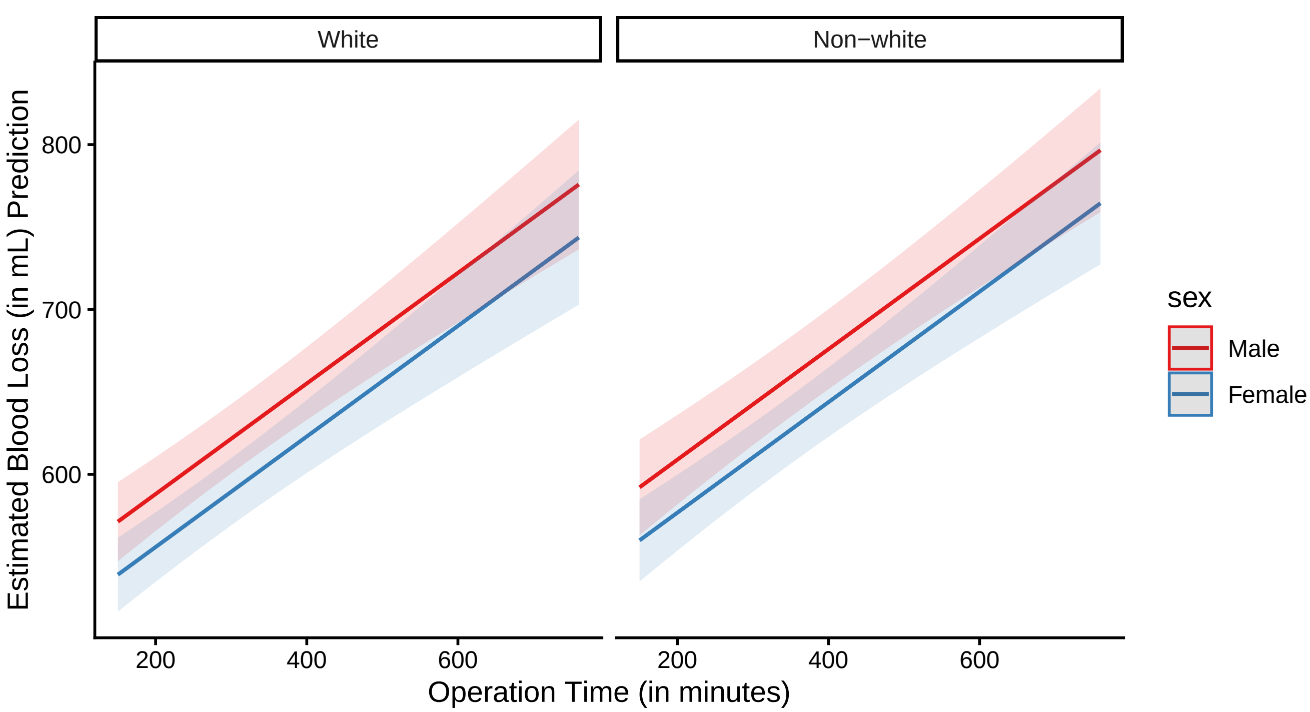
```r
pdf("figures_tables/gamv1_smoothfunc.pdf", width = 8, height = 8)
plot(gam_v2, shade = TRUE, pages = 1)
dev.off()
```

```
## pdf
##   2
```

```
plot(gam_v2, shade = TRUE, pages = 1)
```

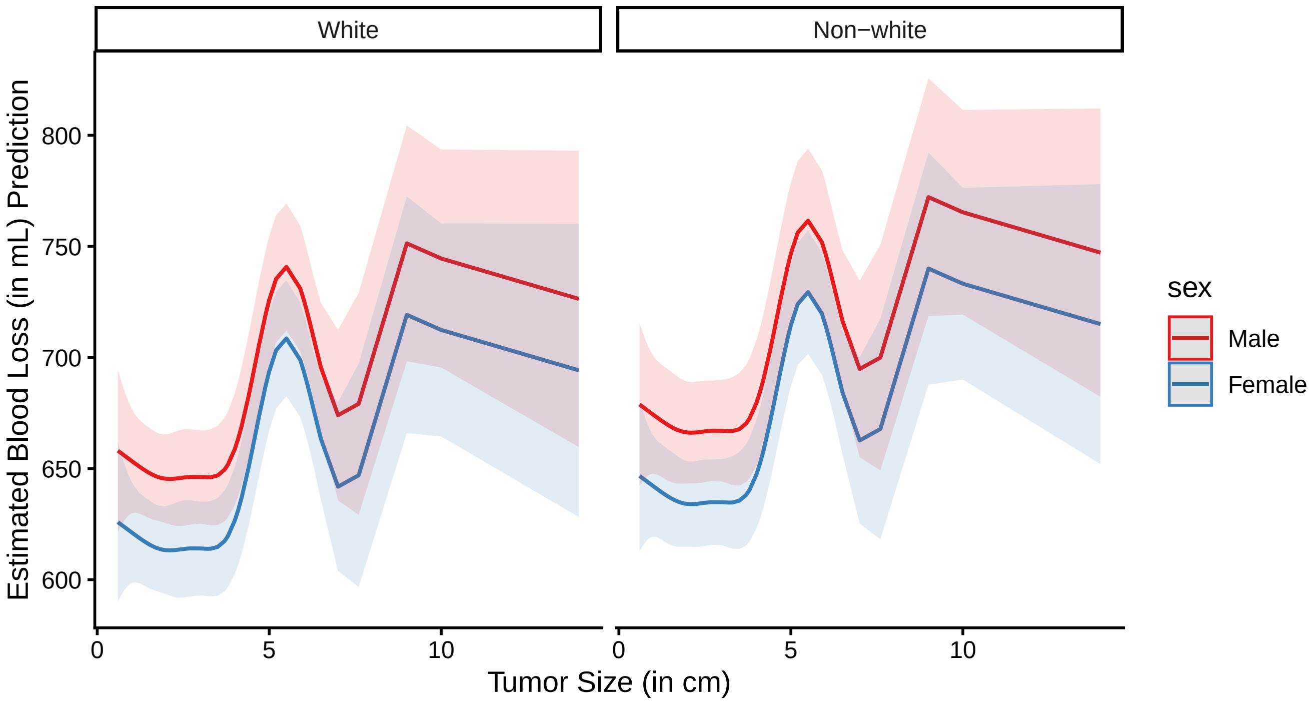



```
plot_model(gam_v2, type = "pred", terms = c("or_time", "sex", "race")) +
  labs(title = "",
       y = "Estimated Blood Loss (in mL) Prediction",
     x = "Operation Time (in minutes)") +
  theme_classic()
```



```
ggsave("figures_tables/ebl_pred_race_gender_operatn_time.pdf", height = 4, width =8)
```

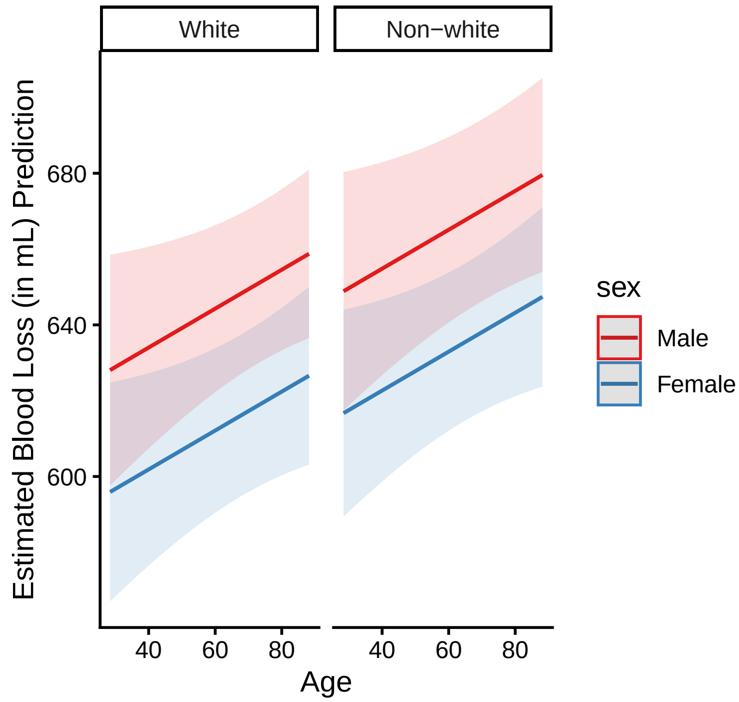
```
plot_model(gam_v2, type = "pred", terms = c("les_size", "sex", "race")) +
  labs(title = "",
       y = "Estimated Blood Loss (in mL) Prediction",
     x = "Tumor Size (in cm)") +
  theme_classic()
```




```
ggsave("figures_tables/ebl_pred_race_gender_liver_test_tumorsize.pdf", height = 4, width =8)
```

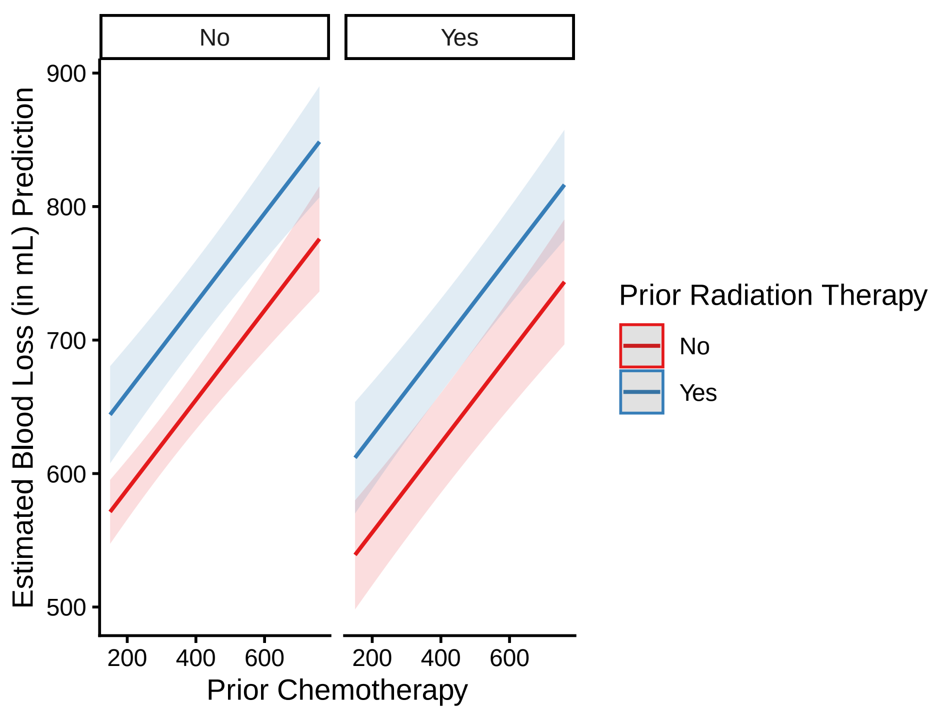
Same trend in men and women across all 4 races

```
plot_model(gam_v2, type = "pred", terms = c("age", "sex", "race"), dodge = 0.6) +
  labs(title = "",
       y = "Estimated Blood Loss (in mL) Prediction",
     x = "Age") +
  theme_classic()
```




```r
ggsave("figures_tables/ebl_pred_race_gender_age.pdf", height = 4, width =8)
```

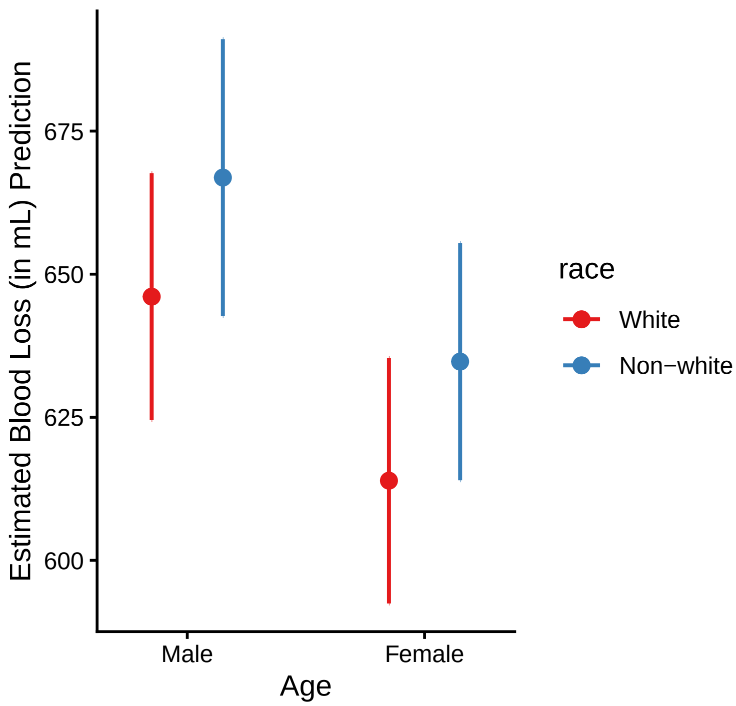
```r
plot_model(gam_v2, type = "pred", terms = c("or_time", "neo_chemo", "neo_xrt"), dodge = 0.6) +
  labs(title = "",
       y = "Estimated Blood Loss (in mL) Prediction",
     x = "Prior Chemotherapy", color = "Prior Radiation Therapy") +
  theme_classic()
```



```
ggsave("figures_tables/ebl_pred_chem_rad_therapy.pdf", height = 4, width =8)
```

Consider that normal bilirubin level is 0.2-1.3. Compare the normal state with degree of increase or decrease from there

```
plot_model(gam_v2, type = "pred", terms = c("sex", "race"), dodge = 0.6) +
  labs(title = "",
       y = "Estimated Blood Loss (in mL) Prediction",
     x = "Age") +
  theme_classic()
```




```
ggsave("figures_tables/ebl_pred_race_gender.pdf", height = 4, width =8)
```

##results table

LM

```r
bc_tbl <- tbl_regression(
  lm_v1,
  intercept= TRUE,
  conf.int = TRUE,
  p.value = TRUE,
  include = everything(),
  tidy_fun = broom.helpers::tidy_parameters,
  estimate_fun = function(x) sprintf("%.5f", x),
    label = list(age ~ "Age (years)",
                 race ~ "Race",
      sex ~ "Sex",
      `liver_test` ~ "Bilirubin Level (in mg/dL)",
      ven_res ~ "Resection + Reconstruction of Major Vasculature",
      neo_chemo ~ "Chemotherapy Prior to Surgery",
      neo_xrt ~ "Radiation Therapy Prior to Surgery",
      `les_size` ~ "Tumor Size (in cm)",
      or_time ~ "Operation Time (in mins)"),
) %>%
  bold_labels() %>%
  modify_header(label ~ "Term")

bc_tbl |>
  as_gt() |>
  gt::tab_options(
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",    # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  gt::tab_style(
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything())) |> gt::gtsave("figures_tables/lm_res_tab.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmp6kWG5S/file11f921adbaec9.html screensh
```

```r
system("pdfcrop figures_tables/lm_res_tab.pdf figures_tables/lm_res_tab.pdf")

bc_tbl |>
  as_kable()
```

Term	Beta	95% CI	p-value
(Intercept)	362.06036	272.55865, 451.56206	<0.001

Term	Beta	95% CI	p-value
Race	NA	NA	
White	NA	NA	
Non-white	10.41707	-6.00953, 26.84368	0.2
Sex	NA	NA	
Male	NA	NA	
Female	-21.61881	-35.96998, -7.26764	0.003
Age (years)	1.92684	0.50912, 3.34456	0.008
Resection + Reconstruction of Major Vasculature	NA	NA	
No	NA	NA	
Yes	27.84709	3.13177, 52.56242	0.027
Chemotherapy Prior to Surgery	NA	NA	
No	NA	NA	
Yes	57.53182	32.66090, 82.40275	<0.001
Radiation Therapy Prior to Surgery	NA	NA	
No	NA	NA	
Yes	-42.28298	-73.74241, -10.82354	0.009
Bilirubin Level (in mg/dL)	1.73034	-0.23989, 3.70057	0.085
Tumor Size (in cm)	10.38624	7.51227, 13.26021	<0.001
Operation Time (in mins)	0.60491	0.37178, 0.83803	<0.001
Age (years) * Operation Time (in mins)	-0.00425	-0.00791, -0.00059	0.023

GAM

```r
bc_tbl <- tbl_regression(
  gam_v2,
  intercept= TRUE,
  conf.int = TRUE,
  p.value = TRUE,
  include = everything(),
  tidy_fun = broom.helpers::tidy_parameters,
  estimate_fun = function(x) sprintf("%.5f", x),
    label = list(age ~ "Age (years)",
                 race ~ "Race",
     sex ~ "Sex",
     `s(liver_test)` ~ "s(Bilirubin Level (in mg/dL))",
     ven_res ~ "Resection + Reconstruction of Major Vasculature",
     neo_chemo ~ "Chemotherapy Prior to Surgery",
     neo_xrt ~ "Radiation Therapy Prior to Surgery",
     `s(les_size)` ~ "s(Tumor Size (in cm))",
     or_time ~ "Operation Time (in mins)"),
) %>%
  bold_labels() %>%
  modify_header(label ~ "Term")
```

```
## i Multinomial models, multi-component models and other groups models have a
##   different underlying structure than the models gtsummary was designed for.
## * Functions designed to work with `tbl_regression()` objects may yield
##   unexpected results.
## i Suppress this message with `?suppressMessages()`.
```

```r
bc_tbl |>
  as_gt() |>
  gt::tab_options(
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",     # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  gt::tab_style(
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything())) |> gt::gtsave("figures_tables/gam_res_tab.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmp6kWG5S/file11f9243a081d0.html screensho
```

```r
system("pdfcrop figures_tables/gam_res_tab.pdf figures_tables/gam_res_tab.pdf")

bc_tbl |>
  as_kable(digits = 3)
```

Group	Term	**Beta**	**95% CI**	**p-value**
conditional	(Intercept)	496.21672	459.71752, 532.71593	<0.001
______	Race	NA	NA	
	White	NA	NA	
	Non-white	20.80744	3.72805, 37.88682	0.017
______	Sex	NA	NA	
	Male	NA	NA	
	Female	-32.15700	-46.39922, -17.91477	<0.001
______	Age (years)	0.51326	0.02479, 1.00173	0.040
______	Resection + Reconstruction of Major Vasculature	NA	NA	
	No	NA	NA	
	Yes	14.78113	-10.58628, 40.14855	0.3
______	Chemotherapy Prior to Surgery	NA	NA	
	No	NA	NA	
	Yes	72.75766	46.66738, 98.84794	<0.001
______	Radiation Therapy Prior to Surgery	NA	NA	
	No	NA	NA	
	Yes	-32.21246	-64.38613, -0.03879	0.050
______	Operation Time (in mins)	0.33531	0.26185, 0.40877	<0.001
smooth_terms	s(Bilirubin Level (in mg/dL))	NA	NA	0.093
______	s(Tumor Size (in cm))	NA	NA	<0.001

```r
## smooth terms part

smooth_df <- as.data.frame(summary(gam_v2)$s.table)
smooth_df$term <- rownames(smooth_df)

smooth_df <- smooth_df[, c("term", "edf", "F", "p-value")]
colnames(smooth_df) <- c("term", "edf", "statistic", "p.value")

smooth_df <-  smooth_df %>% mutate(p.value = case_when(p.value==0~"<0.001", TRUE~as.character(p.value)))
  mutate(term = case_when(term == "s(liver_test)" ~ "s(Bilirubin Level (in mg/dL))",
                          term == "s(les_size)" ~ "s(Tumor Size (in cm))"))


smooth_df %>%
  gt() |>
  gt::tab_options(
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",    # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  gt::tab_style(
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything())) |>
  tab_header(title = "Smooth Terms") |>
  gt::gtsave("figures_tables/gam_smooth_eff.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmp6kWG5S/file11f92290f3578.html screensh
```

```r
system("pdfcrop figures_tables/gam_smooth_eff.pdf figures_tables/gam_smooth_eff.pdf")
```